Pattern making using looping and file handling (coding screenshots)

This document contains different possible combinations of right-angled triangle, left angled triangle and pyramid using numbers, alphabets in order and alphabets in a name. The output txt file has been uploaded in the Github repository and the name of the file is: **ak. txt pattern making_file handling_output**

[114]:
```
def ratnum():
    f=open('ak.txt','w+')
    f.write('Right angle triangle using number(rows)''\n''--------------------------------------''\n')
    for i in range(1,6):
        for j in range(0,i):
            f.write(str(i))
        f.write('\n')
    f.close()
ratnum()
```

[13]:
```
import os
print(os.getcwd())
```

```
C:\Users\D E L L\Desktop\python Gen AI
```

[115]:
```
def ratnumcol():
    f=open('ak.txt','a+')
    f.write('Right angle triangle using numbers(column)''\n''--------------------------------------------''\n')
    for i in range(1,6):
        for j in range(0,i):
            f.write(str(j))
        f.write('\n')
    f.close()
ratnumcol()
```

[116]:
```
def ratstar():
    f=open('ak.txt','a+')
    f.write('Right angle triangle using *(rows)''\n''-------------------------------------------''\n')
    for i in range(1,6):
        for j in range(0,i):
            f.write(('*'))
        f.write('\n')
    f.close()
ratstar()
```

[117]:
```
def iratnumrow():
    f=open('ak.txt','a+')
    f.write('Inverse right angled triangle using numbers(row)''\n''------------------------------------------------------''\n')
    for i in range(6,0,-1):
        for j in range(0,i):
            f.write(str(i))
        f.write('\n')
    f.close()
iratnumrow()
```

[118]:
```
def iratnumcol():
    f=open('ak.txt','a+')
    f.write('Inverse right angled triangle using numbers(column)''\n''------------------------------------------------------''\n')
    for i in range(6,0,-1):
        for j in range(0,i):
            f.write(str(j))
        f.write('\n')
    f.close()
iratnumcol()
```

[119]:
```
def ratupperrow():
    name= 'Akshara'.upper()
    f=open('ak.txt','a+')
    f.write('Right angled triangle in upper case(row)''\n''-----------------------------------------------''\n')
    for i in range(len(name)):
        for j in range(0,i+1):
            f.write(name[i])
        f.write('\n')
    f.close()
ratupperrow()
```

[120]:

```python
def ratlowerrow():
    name='akshara'
    f=open('ak.txt','a+')
    f.write('Right angled triangle in lower case(row)''\n''-------------------------------------------------------''\n')
    for i in range(len(name)):
        for j in range(0,i+1):
            f.write(name[i])
        f.write('\n')
    f.close()
ratlowerrow()
```

[121]:

```python
def iratuppercol():
    name= 'akshara'.upper()
    f=open('ak.txt','a+')
    f.write('Inverse right angle triangle uppercase(column)''\n''--------------------------------------------------''\n')
    for i in range(len(name),-1,-1):
        for j in range(0,i-1):
            f.write(name[j])
        f.write('\n')
    f.close()
iratuppercol()
```

[122]:

```python
def iratlowercol():
    name= 'akshara'.lower()
    f=open('ak.txt','a+')
    f.write('Inverse right angle triangle lowercase(column)''\n''--------------------------------------------------''\n')
    for i in range(len(name),-1,-1):
        for j in range(0,i):
            f.write(name[j])
        f.write('\n')
    f.close()
iratlowercol()
```

[123]:

```python
def iratlowerrow():
    name= 'akshara'.lower()
    f=open('ak.txt','a+')
    f.write('Inverse right angle triangle lowercase(row)''\n''--------------------------------------------------''\n')
    for i in range(len(name)):
        for j in range(0,i+1):
            f.write(name[i])
        f.write('\n')
    f.close()
iratlowerrow()
```

[124]:

```python
def ratalphaupperrow():
    f=open('ak.txt','a+')
    f.write('Right angle triangle using upper case alphabet(row)''\n''----------------------------------------------------------------
    for i in range(0,6):
        for j in range(0,i):
            f.write(chr(i+64))
        f.write('\n')
    f.close()
ratalphaupperrow()
```

[125]:

```python
def ratalphalowerrow():
    f=open('ak.txt','a+')
    f.write('Right angle triangle suing lower case alphabet(row)''\n''----------------------------------------------------------------
    for i in range(0,6):
        for j in range(0,i):
            f.write(chr(i+64+32))
        f.write('\n')
    f.close()
ratalphalowerrow()
```

[126]:
```python
def ratalphauppercol():
    f=open('ak.txt','a+')
    f.write('Right angle triangle using upper case alphabet(col)''\n''------------------------------------------------------------------''\n')
    for i in range(0,6):
        for j in range(0,i+1):
            f.write(chr(j+65))
        f.write('\n')
    f.close()
ratalphauppercol()
```

[127]:
```python
def ratalphalowercol():
    f=open('ak.txt','a+')
    f.write('Right angle triangle suing lower case alphabet(col)''\n''-------------------------------------------------------------------------''\n')
    for i in range(0,6):
        for j in range(0,i+1):
            f.write(chr(i+65+32))
        f.write('\n')
    f.close()
ratalphalowercol()
```

[128]:
```python
def iratalphaupperrow():
    f=open('ak.txt','a+')
    f.write('Inverse right angle triangle using upper case alphabet(row)''\n''------------------------------------------------------------------''\n')
    for i in range(0,6):
        for j in range(7,i,-1):
            f.write(chr(i+65))
        f.write('\n')
    f.close()
iratalphaupperrow()
```

[129]:
```python
def latnumrow():
    f=open('ak.txt','a+')
    f.write('\n''Left angled triangle using number(row)''\n''--------------------------------------------''\n')
    for i in range(0,6):
        for k in range(6,i,-1):
            f.write(' ')
        for j in range(0,i):
            f.write(str(i))
        f.write('\n')
    f.close()
latnumrow()
```

[130]:
```python
def latnumcol():
    f=open('ak.txt','a+')
    f.write('\n''left angle triangle with numbers(column)''\n''--------------------------------------------''\n')
    for i in range(0,7):
        for k in range(7,i,-1):
            f.write(' ')
        for j in range(0,i):
            f.write(str(j))
        f.write('\n')
    f.close()
latnumcol()
```

[131]:
```python
def latstar():
    f=open('ak.txt','a+')
    f.write('\n''left angle triangle with star''\n')
    for i in range(0,6):
        for k in range(6,i,-1):
            f.write(' ')
        for j in range(0,i):
            f.write('*')
        f.write('\n')
    f.close()
latstar()
```

[132]:
```
def latlowrow():
    name='AKSHARA'.lower()
    f=open('ak.txt','a+')
    f.write('\n''Left angle triangle using name(row)''\n''--------------------------------------------''\n')
    for i in range(len(name)):
        for k in range(7,i,-1):
            f.write(' ')
        for j in range(0,i+1):
            f.write(name[i])
        f.write('\n')
    f.close()
latlowrow()
```

[133]:
```
def latupprow():
    name= 'akshara'.upper()
    f=open('ak.txt','a+')
    f.write('\n''Left angle triangle using name(upper-row)''\n''--------------------------------------------------''\n')
    for i in range(len(name)):
        for k in range(7,i,-1):
            f.write(' ')
        for j in range(0,i+1):
            f.write(name[i])
        f.write('\n')
    f.close()
latupprow()
```

[134]:
```
def latlowcol():
    name='AKSHARA'.lower()
    f=open('ak.txt','a+')
    f.write('\n''Left angle triangle using name(col)''\n''----------------------------------------------''\n')
    for i in range(len(name)):
        for k in range(7,i,-1):
            f.write(' ')
        for j in range(0,i+1):
            f.write(name[j])
        f.write('\n')
    f.close()
latlowcol()
```

[135]:
```
def latuppcol():
    name='akshara'.upper()
    f=open('ak.txt','a+')
    f.write('\n' 'Left angle triangle using name(upper-column)''\n''-----------------------------------------------------''\n')
    for i in range(len(name)):
        for k in range(7,i,-1):
            f.write(' ')
        for j in range(0,i+1):
            f.write(name[j])
        f.write('\n')
    f.close()
latuppcol()
```

[136]:
```
def latalphaupperrow():
    f=open('ak.txt','a+')
    f.write('\n''left angle triangle with upper case alphabets(row)''\n''-------------------------------------------------''\n')
    for i in range(0,5):
        for k in range(5,i,-1):
            f.write(' ')
        for j in range(0,i):
            f.write(str(1+64))
        f.write('\n')
    f.close()
latalphaupperrow()
```

[137]:
```python
def ilatnumrow():
    f=open('ak.txt','a+')
    f.write('\n''Inverse left angled triangle using number(row)''\n''--------------------------------------------''\n')
    for i in range(6,0,-1):
        for k in range(6,i,-1):
            f.write(' ')
        for j in range(0,i):
            f.write(str(i))
        f.write('\n')
    f.close()
ilatnumrow()
```

[138]:
```python
def ilatnumcol():
    f=open('ak.txt','a+')
    f.write('\n''Inverse left angled triangle using number(column)''\n''--------------------------------------------''\n')
    for i in range(6,0,-1):
        for k in range(6,i,-1):
            f.write(' ')
        for j in range(0,i-1):
            f.write(str(j))
        f.write('\n')
    f.close()
ilatnumcol()
```

[139]:
```python
def ilatstar():
    f=open('ak.txt','a+')
    f.write('\n''inverse left angle triangle with star''\n')
    for i in range(6,0,-1):
        for k in range(6,i,-1):
            f.write(' ')
        for j in range(0,i):
            f.write('*')
        f.write('\n')
    f.close()
ilatstar()
```

[140]:
```python
def ilatupperrow():
    name='akshara'.upper()
    f=open('ak.txt','a+')
    f.write('\n''inverse left angle triangle with name upper case(row)''\n''---------------------------------------------------------''\n')
    for i in range(len(name)-1,-1,-1):
        for k in range(6,0,-1):
            f.write(' ')
        for j in range(0,i+1):
            f.write(str(name[i]))
        f.write('\n')
    f.close()
ilatupperrow()
```

[141]:
```python
def ratnumcol():
    f=open('ak.txt','a+')
    f.write('Right angle triangle using numbers(column)''\n''------------------------------------------''\n')
    for i in range(1,6):# range index:1,2,3,4,5
        for j in range(0,i):
            f.write(str(j)) #print j as string
        f.write('\n') # for new line
    f.close() #closing the opened ak.txt file
ratnumcol() #calling the function
```

[142]:
```python
def ratstar():
    f=open('ak.txt','a+')
    f.write('Right angle triangle using *(rows)''\n''------------------------------------------''\n')
    for i in range(1,6): # range:1,2,3,4,5
        for j in range(0,i):
            f.write(('*')) #to print star
        f.write('\n') # for new line
    f.close() #closing the opened ak.txt file
ratstar() #calling the function
```

[142]:
```
def ratstar():
    f=open('ak.txt','a+')
    f.write('Right angle triangle using *(rows)''\n''-------------------------------------------''\n')
    for i in range(1,6): # range:1,2,3,4,5
        for j in range(0,i):
            f.write(('*')) #to print star
        f.write('\n') # for new line
    f.close() #closing the opened ak.txt file
ratstar() #calling the function
```

[143]:
```
def iratnumrow():
    f=open('ak.txt','a+')
    f.write('Inverse right angled triangle using numbers(row)''\n''----------------------------------------------------''\n')
    for i in range(6,0,-1): #decrement -1 as we need the inverse range from range index 6,5,4,3,2,1.
        for j in range(0,i):
            f.write(str(i)) #print i as string
        f.write('\n') # for new line
    f.close()#closing the opened ak.txt file
iratnumrow() #calling the function
```

[144]:
```
def pyramidnumrow():
    f=open('ak.txt','a+')
    f.write('\n''pyramid using numbers(row)''\n''----------------------\n')
    for i in range(0,6):
        for k in range(6,i,-1):
            f.write(' ')
        for j in range(0,i):
            f.write(str(i)+' ')
        f.write('\n')
    f.close()

pyramidnumrow()
```

[145]:
```
def iratnumrow():
    f=open('ak.txt','a+')
    f.write('Inverse right angled triangle using numbers(row)''\n''----------------------------------------------------''\n')
    for i in range(6,0,-1): #decrement -1 as we need the inverse range from range index 6,5,4,3,2,1.
        for j in range(0,i):
            f.write(str(i)) #print i as string
        f.write('\n') # for new line
    f.close()#closing the opened ak.txt file
iratnumrow() #calling the function
```

[146]:
```
def pyramidnumcol():
    f=open('ak.txt','a+')
    f.write('\n''pyramid using numbers(col)''\n''----------------------\n')
    for i in range(0,6):
        for k in range(6,i,-1):
            f.write(' ')
        for j in range(0,i):
            f.write(str(j+1)+' ')
        f.write('\n')
    f.close()

pyramidnumcol()
```

[147]:
```
def pyramidstar():
    f=open('ak.txt','a+')
    f.write('\n''pyramid using star''\n''----------------------\n')
    for i in range(0,6):
        for k in range(6,i,-1):
            f.write(' ')
        for j in range(0,i):
            f.write(str('*') +' ')
        f.write('\n')
    f.close()

pyramidstar()
```

[148]:
```
def ipyramidstar():
    f=open('ak.txt','a+')
    f.write('\n''inverse pyramid using star''\n''----------------------\n')
    for i in range(6,0,-1):
        for k in range(6,i,-1):
            f.write(' ')
        for j in range(0,i):
            f.write(str('*') +' ')
        f.write('\n')
    f.close()

ipyramidstar()
```

[149]:
```
def ipyramidnumcol():
    f=open('ak.txt','a+')
    f.write('\n''inverse pyramid using numbers(col)''\n''----------------------\n')
    for i in range(6,0,-1):
        for k in range(6,i,-1):
            f.write(' ')
        for j in range(0,i):
            f.write(str(j+1)+' ')
        f.write('\n')
    f.close()

ipyramidnumcol()
```

[150]:
```
def ipyramidnumrow():
    f=open('ak.txt','a+')
    f.write('\n''inverse pyramid using numbers(row)''\n''----------------------\n')
    for i in range(6,0,-1):
        for k in range(6,i,-1):
            f.write(' ')
        for j in range(0,i):
            f.write(str(i)+' ')
        f.write('\n')
    f.close()

ipyramidnumrow()
```

[151]:
```
def pyramidrow():
    name='akshara'.upper()
    f=open('ak.txt','a+')
    f.write('\n''pyramid using name(row)''\n''----------------------\n')
    for i in range(len(name)):
        for k in range(7,i,-1):
            f.write(' ')
        for j in range(0,i+1):
            f.write(name[i]+' ')
        f.write('\n')
    f.close()

pyramidrow()
```

[152]:
```
def pyramidcol():
    name='akshara'.upper()
    f=open('ak.txt','a+')
    f.write('\n''pyramid using name(col)''\n''----------------------\n')
    for i in range(len(name)):
        for k in range(7,i,-1):
            f.write(' ')
        for j in range(0,i+1):
            f.write(name[j]+' ')
        f.write('\n')
    f.close()

pyramidcol()
```

[153]:
```
def pyramidalpharow():
    f=open('ak.txt','a+')
    f.write('\n''pyramid using alpha(row)''\n''-----------------------\n')
    for i in range(0,6):
        for k in range(7,i,-1):
            f.write(' ')
        for j in range(0,i+1):
            f.write(chr(i+65)+' ')
        f.write('\n')
    f.close()

pyramidalpharow()
```

[154]:
```
def pyramidalphacol():
    f=open('ak.txt','a+')
    f.write('\n''pyramid using alpha(col)''\n''-----------------------\n')
    for i in range(0,6):
        for k in range(7,i,-1):
            f.write(' ')
        for j in range(0,i+1):
            f.write(chr(j+65)+' ')
        f.write('\n')
    f.close()

pyramidalphacol()
```

[155]:
```
def ipyramidrow():
    name='akshara'.upper()
    f=open('ak.txt','a+')
    f.write('\n''inverse pyramid using name(row)''\n''-----------------------\n')
    for i in range(len(name),0,-1):
        for k in range(7,i,-1):
            f.write(' ')
        for j in range(0,i):
            f.write(name[i-1]+' ')
        f.write('\n')
    f.close()

ipyramidrow()
```

[156]:
```
def ipyramidcol():
    name='akshara'.lower()
    f=open('ak.txt','a+')
    f.write('\n''inverse pyramid using name(col)''\n''-----------------------\n')
    for i in range(len(name),0,-1):
        for k in range(7,i,-1):
            f.write(' ')
        for j in range(0,i):
            f.write(name[j-1]+' ')
        f.write('\n')
    f.close()

ipyramidcol()
```

[157]:
```
def ipyramidalphacol():
    f=open('ak.txt','a+')
    f.write('\n''inverse pyramid using alpha(col)''\n''-----------------------\n')
    for i in range(6,0,-1):
        for k in range(7,i,-1):
            f.write(' ')
        for j in range(0,i):
            f.write(chr(j+65)+' ')
        f.write('\n')
    f.close()

ipyramidalphacol()
```

[158]:
```
def ipyramidalpharow():
    f=open('ak.txt','a+')
    f.write('\n''inverse pyramid using alpha(row)''\n''-----------------------\n')
    for i in range(6,0,-1):
        for k in range(7,i,-1):
            f.write(' ')
        for j in range(0,i):
            f.write(chr(i+64)+' ')
        f.write('\n')
    f.close()

ipyramidalpharow()
```

[ ]: